Chapter 6

Constructing Analytical Energy Functions for Transient Stability Models

6.1 INTRODUCTION

The task of constructing an energy function for a (postfault) transient stability model is essential to direct methods. The role of the energy function is to make feasible a direct determination of whether a given point (such as the initial point of a postfault power system) lies inside the stability region of a postfault stable equilibrium point (SEP) without performing numerical integration. An energy function is an extension of the Lyapunov function in that it must satisfy the three conditions described in the previous chapter. A Lyapunov function may not be an energy function.

Two different classes of power system models for direct transient stability analysis have been proposed: network-reduction models and network-preserving models. Traditionally, direct methods have been developed for the network-reduction models where all the loads are modeled as constant impedances and the entire network representation is reduced to the generator internal buses. Network-preserving models were proposed in the 1980s to overcome some of the shortcomings of network-reduction models. In each class of model, a model is termed lossless if the nonzero transfer conductance terms are neglected; otherwise, it is termed lossy, such as lossy network-reduction models and lossy network-preserving models.

Practical power system stability models are lossy, where the losses are either from the transmission system and loads or from the transfer conductances in the reduced system admittance matrix after the elimination of load buses. Considerable efforts have concentrated on the construction of analytical energy functions for lossy power system stability models (Guadra, 1975; Henner, 1976; Machias, 1986; Pai and Murthy, 1973). These efforts, however, have been in vain.

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On the other hand, most of the existing energy functions for power systems with losses are based on unjustified approximations (Athay et al., 1979; Hartman, 1973; Kakimoto et al., 1978; Pai and Varwandkar, 1977; Uemura et al., 1972). These functions are either not well-defined or the derivative of these functions along system trajectories is not negative. Thus, the practical merits of these energy functions for the stability analysis of power systems with losses are unclear. Indeed, it has been shown that the energy function in Uemura et al. (1972) may lead to an estimate of critical clearing time higher than its actual value obtained through step-by-step methods.

Consequently, these efforts raise the issue of the existence of energy functions for power systems with losses. Regarding this issue, the existence of local Lyapunov functions for power systems with transfer conductances has been shown in Caprio (1986), Kwatny et al. (1985), Narasimhamurthi and Musavi (1984), and Saeki et al. (1983). Unfortunately, these local Lyapunov functions only serve to determine the stability property of an equilibrium point and cannot be directly applied to the determination of the stability region. As a result, these local Lyapunov functions cannot be employed in direct methods for transient stability analysis.

The issue therefore has been only partly resolved: there is a local energy function for power systems with losses. However, does a global energy function (or Lyapunov function) for lossy power system stability models exist? This question was addressed in Chiang (1989) by demonstrating the nonexistence of a general energy function for power systems with losses. While this may be a disappointing message, it has also been shown that under certain conditions, there is an energy function (defined in a compact set of the state space) that can be used in direct methods for the stability analysis of power systems with certain losses. One implication of these results is that any general procedure attempting to construct an analytical energy function for a power system with losses must include a step that checks its existence. These theoretical results confirm the current practice of using numerical energy functions for direct transient stability analysis.

This chapter is focused on the methodology of constructing (analytical) energy functions for lossless power system stability models. In addition, a theoretical study of the nonexistence of energy functions for lossy power system stability models will be presented. The construction of (numerical) energy functions for lossy power system stability models will be presented in the next chapter.

6.2 ENERGY FUNCTIONS FOR LOSSLESS NETWORK-REDUCTION MODELS

Many existing lossless network-reduction transient stability models can be written in the following compact form (Chu and Chiang, 1999):

$$
\begin{aligned}
T\dot{x} &= -\frac{\partial U}{\partial x}(x, y) \\
\dot{y} &= z \\
M\dot{z} &= -Dz - \frac{\partial U}{\partial y}(x, y),
\end{aligned}
\tag{6.1}
$$

where $x \in R^n$, y, $z \in R^m$, T, M, and D are positive diagonal matrices. $U(x, y)$ is a smooth function. A sufficient condition for the existence of an energy function for this compact representation is the following (Chu and Chiang, 1999):

$$W(x, y, z) = K(z) + U(x, y) = \frac{1}{2} z^T Mz + U(x, y), \tag{6.2}$$

where the function W: $R^{n+2m} \to R$. If, along every nontrivial trajectory $(x(t), y(t), z(t))$ with a bounded function value $W(x, y, z)$, $x(t)$ is also bounded, then $W(x, y, z)$ is an energy function for the system.

These classical lossless, network-reduction transient stability models are seldom used in practical applications because of the underlying unwarranted simplifications in the formulation of generators and loads. This type of model is known to have the following shortcomings: (1) it precludes the models for nonlinear load behaviors; (2) reduction of the network leads to loss of network topology and hence precludes the study of energy shifts among different components of the network; (3) neglect of transfer conductances leads to a bias in stability assessments of unknown magnitude and direction (overestimates or underestimates). Structure-preserving models have been proposed to overcome some of the shortcomings of network-reduction models. We discuss in the next section a general procedure to derive analytical energy functions for network-preserving transient stability models.

6.3 ENERGY FUNCTIONS FOR LOSSLESS STRUCTURE-PRESERVING MODELS

We construct analytical energy functions for network-preserving lossless transient stability models. Several existing network-preserving transient stability models can be rewritten as a set of general differential and algebraic equations (DAEs) of the following compact form (Bergen and Hill, 1981; Bergen et al., 1986; Chu and Chiang, 2005; Padiyar and Ghosh, 1989; Padiyar and Sastry, 1987):

$$\begin{aligned}
0 &= -\frac{\partial}{\partial u} U(u, w, x, y) \\
0 &= -\frac{\partial}{\partial w} U(u, w, x, y) \\
T\dot{x} &= -\frac{\partial}{\partial x} U(u, w, x, y) \\
\dot{y} &= z \\
M\dot{z} &= -Dz - \frac{\partial}{\partial y} U(u, w, x, y),
\end{aligned} \tag{6.3}$$

where $u \in R^k$ and $w \in R^l$ are instantaneous variables and $x \in R^n$, $y \in R^n$, and $z \in R^n$ are state variables. T is a positive definite matrix, and M and D are diagonal positive definite matrices. Here, differential equations describe generator and/or load dynamics, while algebraic equations express the power flow equations at each bus. Also, the function $U(u, w, x, y)$ satisfies the following conditions:

(D1) $\nabla U(u, w, x, y) = \nabla U(u_1 + 2m_1 \pi, \ldots, u_k + 2m_k \pi, w, x, y_1 + 2l\pi, \ldots, y_n + 2l_n \pi)$ for all $m_i, l_j \in Z$.

(D2) For $i = 1, \ldots, n$, $k = 1, \ldots, n$, and $j = 1, \ldots, m$, there exist four polynomials with positive coefficients $p_{1j}(u_1, \ldots, u_k, x_1, \ldots, x_m)$, $p_{2ik}(u_1, \ldots, u_k, x_1, \ldots, x_m)$, $p_{3ij}(u_1, \ldots, u_k, x_1, \ldots, x_m)$, and $p_{4ij}(u_1, \ldots, u_k, x_1, \ldots, x_m)$, such that

$$\left|\frac{\partial}{\partial y_i} U(u, w, x, y)\right| \le p_{1i}\left(|u_1|, \ldots, |u_k|, |x_1|, \ldots, |x_m|\right),$$

$$\left|\frac{\partial}{\partial^2 y_i y_k} U(u, w, x, y)\right| \le p_{2ik}\left(|u_1|, \ldots, |u_k|, |x_1|, \ldots, |x_m|\right),$$

$$\left|\frac{\partial}{\partial w_j \partial y_i} U(u, w, x, y)\right| \le p_{3ij}\left(|u_1|, \ldots, |u_k|, |x_1|, \ldots, |x_m|\right), \text{ and}$$

$$\left|\frac{\partial}{\partial x_j \partial y_i} U(u, w, x, y)\right| \le p_{4ij}\left(|u_1|, \ldots, |u_k|, |x_1|, \ldots, |x_m|\right),$$

respectively.

Due to their intricate and complicated dynamics near singular surfaces, DAE systems are usually difficult to analyze. To avoid an awkward analysis on the DAE representation, various methods have been presented in the literature. Here, we will treat the algebraic equations as singularly perturbed differential equations (Chiang et al., 1994). The application of the singularly perturbed systems in network-preserving power system models was initiated by Sastry, Desour, and Varaiya (Sastry and Desoer, 1981; Sastry and Varaiya, 1980) and was considered later by Arapostathis et al. (1982), Bergen et al. (1986), De Marco and Bergen (1984, 1987), Chiang et al. (1992), and Zou et al. (2003).

With the aid of singularly perturbed systems, the compact representation of the network-preserving, lossless, transient stability model becomes

$$\begin{aligned}
\varepsilon_1 \dot{u} &= -\frac{\partial}{\partial u} U(u, w, x, y) \\
\varepsilon_1 \dot{w} &= -\frac{\partial}{\partial w} U(u, w, x, y) \\
T\dot{x} &= -\frac{\partial}{\partial x} U(u, w, x, y) \\
\dot{y} &= z \\
M\dot{z} &= -Dz - \frac{\partial}{\partial y} U(u, w, x, y),
\end{aligned} \tag{6.4}$$

where ε_1 and ε_2 are sufficiently small positive numbers.

We next construct an analytical energy function for the lossless network-preserving transient stability model.

Theorem 6.1: Existence of an Analytical Energy Function

For the compact representation of the singularly perturbed network-preserving power system model (Eq. 6.4), consider the function W: $R^{5n} \to R$:

$$W(u, w, x, y, z) = K(z) + U(u, w, x, y) = \frac{1}{2} z^T M z + U(u, w, x, y). \tag{6.5}$$

Suppose that along every nontrivial trajectory ($u(t)$, $w(t)$, $x(t)$, $y(t)$, $z(t)$) of the system (Eq. 6.4) with bounded $W(. , . , . , . , .)$, ($u(t)$, $w(t)$, $x(t)$) is also bounded. Then, $W(u, w, x, y, z)$ is an energy function for the system (Eq. 6.4).

Proof: We will check conditions (i)–(iii) of the energy function:

(i) Differentiating $W(u(t), w(t), x(t), y(t), z(t))$ along the trajectory yields the following:

$$\begin{aligned}\dot{W}(u(t), w(t), x(t), y(t), z(t)) &= \frac{\partial W^T}{\partial u}\dot{u} + \frac{\partial W^T}{\partial w}\dot{w} + \frac{\partial W^T}{\partial x}\dot{x} + \frac{\partial W^T}{\partial y}\dot{y} + \frac{\partial W^T}{\partial z}\dot{z} \\ &= -\frac{\partial U^T}{\partial u}\varepsilon_1^{-1}\frac{\partial U}{\partial u} - \frac{\partial U^T}{\partial w}\varepsilon_2^{-1}\frac{\partial U}{\partial w} - \frac{\partial U^T}{\partial x}T^{-1}\frac{\partial U}{\partial x} \\ &= -z^T D z \le 0.\end{aligned}$$

This inequality shows that condition (i) of the energy function is satisfied.

(ii) Suppose that there is an interval $t \in [t_1, t_2]$ such that $\dot{W}(u(t), w(t), x(t), y(t), z(t)) = 0$. Hence, it follows that $z(t) = 0$ and $\dot{u}(t) = \dot{w}(t) = \dot{x}(t) = 0$ for $t \in [t_1, t_2]$. But this also implies that $y(t)$ is a constant, which implies that the system is at an equilibrium point. Hence, condition (ii) of the energy function also holds.

(iii) Because $u(t)$, $w(t)$, and $x(t)$ are bounded along the nontrivial trajectory with bounded $W(. , . , . , . , .)$, only components $y(t)$ and $z(t)$ will be examined for condition (iii). We then apply Barbalat's lemma and the covering map to conclude that $u(t)$, $w(t)$, $x(t)$, $y(t)$, and $z(t)$ are bounded. From conditions (i)–(iii) we conclude that $W(u(t), w(t), x(t), y(t), z(t))$ is indeed an energy function. This completes the proof.

For an illustrational purpose, we next derive an analytical energy function for the two-axis generator with first-order exciter dynamics and a dynamic load model with the following characteristics:

1. Each generator is modeled as a two-axis generator plus a first-order exciter system.

2. Each load demand is a hybrid of constant susceptance, constant complex power, and a frequency-dependent dynamic load model.
3. For an internal generator bus, the two-axis generator model is written as follows: For $i = 1, \ldots, n$,

$$\begin{aligned}
\dot{\delta}_i &= \omega_i \\
M_i\dot{\omega}_i &= -D_i\omega_i + P_{mi} - P_{ei} \\
T_{doi}E'_{qi} &= -\frac{x_{di}}{x'_{di}}E'_{qi} + \frac{x_{di} - x'_{di}}{x'_{di}}V_i\cos(\delta_i - \theta_i) + E_{fi} \\
T_{qoi}E'_{di} &= -\frac{x_{qi}}{x'_{qi}}E'_{di} - \frac{x_{qi} - x'_{qi}}{x'_{qi}}V_i\sin(\delta_i - \theta_i),
\end{aligned} \tag{6.6}$$

where

$$P_{ei} = \frac{1}{x'_{di}}E'_{qi}V_i\sin(\delta_i - \theta_i) + \frac{1}{x'_{qi}}E'_{di}V_i\cos(\delta_i - \theta_i) + \frac{x'_{di} - x'_{qi}}{2x'_{di}x'_{qi}}V_i^2\sin[2(\delta_i - \theta_i)].$$

4. Exciter dynamics: various exciter dynamic models have been proposed in the past. For simplicity and to avoid the nonsmooth dynamics of hard limits in the detailed exciter models, the following first-order approximation exciter dynamic is considered:

$$T_{vi}\dot{E}_{fi} = -E_{fi} - \mu_i k_i V_i\cos(\delta_i - \theta_i) + l_i. \tag{6.7}$$

5. For an external generator bus, the power flow equations at each generator can be described in the following equations. For $i = 1, \ldots, n$,

$$\begin{aligned}
0 = &\sum_{j\neq i}^{n+1} V_iV_j\left(B_{ij}\sin(\theta_i - \theta_j)\right) + G_{ij}\cos(\theta_i - \theta_j) \\
&+ \sum_{l=n+2}^{n+m+1} V_iV_l\left(B_{il}\sin(\theta_i - \theta_l) + G_{il}\cos(\theta_i - \theta_l)\right) \\
&- \frac{1}{x'_{di}}E'_{qi}V_i\sin(\delta_i - \theta_i) - \frac{1}{x'_{di}}E'_{di}V_i\cos(\delta_i - \theta_i) \\
&- \frac{x'_{di} - x'_{qi}}{2x'_{di}x'_{qi}}V_i^2\sin[2(\delta_i - \theta_i)] \text{ and}
\end{aligned} \tag{6.8}$$

$$\begin{aligned}
0 = &\frac{V_i^2}{x'_{di}} - \frac{1}{x'_{di}}E'_{qi}V_i\cos(\delta_i - \theta_i) - \frac{1}{x'_{qi}}E'_{di}V_i\sin(\theta_i - \delta_i) \\
&- \frac{x'_{di} - x'_{qi}}{2x'_{di}x'_{qi}}V_i^2[\cos 2(\delta_i - \theta_i) - 1] \\
&- \sum_{j=1}^{n+1} V_iV_j\left(B_{ij}\cos(\theta_i - \theta_j)\right) + G_{ij}\sin(\theta_i - \theta_j) \\
&- \sum_{l=n+2}^{n+m+1} V_iV_l\left(B_{il}\cos(\theta_i - \theta_l) + G_{il}\sin(\theta_i - \theta_l)\right).
\end{aligned} \tag{6.9}$$

6. Load bus: because the portion of the constant impedance load is incorporated into the network admittance matrix Y, the load demand at each load bus is a hybrid combination of the following three terms: (1) frequency-dependent loads—$D_k\dot{\theta}_k$; (2) constant power $P_k^d + JQ_k^d(V_k)$; and (3) constant current injection $I_{Lk}\angle\phi_k$, where both I_{Lk} and ϕ_k are constants. The bus voltage angle θ will be changed during the transient stage. So, the power factor of the constant current injection will not remain a constant during the transient response. Hence, the complex power demand at each node is of the form $P_k^d + V_k I_{Lk}\cos(\theta_k - \phi_k) + j\left[Q_k^d(V_k) + V_k I_{Lk}\sin(\theta_k - \phi_k)\right]$. The power flow equations at the load bus can be represented in the following equations: for $k = n + 2, \ldots, n + m + 1$,

$$P_k^d + V_k I_{Lk}\cos(\theta_k - \phi_k) - D_k\dot{\theta}_k = \sum_{j=1}^{n+m+1} V_k V_j\left(B_{kj}\sin(\theta_k - \theta_j) + G_{kj}\cos(\theta_k - \theta_j)\right)$$

$$Q_k^d(V_k) + V_k I_{Lk}\sin(\theta_k - \phi_k) = -\sum_{j=1}^{n+m+1} V_k V_j\left(B_{kj}\cos(\theta_k - \theta_j) + G_{kj}\sin(\theta_k - \theta_j)\right). \tag{6.10}$$

We consider the function (Chu and Chiang, 2005)

$$U(\delta, \theta, V, E_q', I_L) = \sum_{i=1}^{14} U_i, \tag{6.11}$$

where

$$U_1 = -\sum_{i=1}^{n} P_{mi}\delta_i,\ U_2 = -\sum_{i<j}^{n+1} V_i V_j B_{ij}\cos(\theta_i - \theta_j),\ U_3 = -\sum_{i=1}^{n+1}\sum_{k=n+2}^{n+m+1} V_i V_k B_{ik}\cos(\theta_i - \theta_k)$$

$$U_4 = -\sum_{k=n+2}^{n+m+1} P_k^d\delta_k,\ U_5 = -\sum_{k<1}^{n+m+1} V_k V_l B_{kl}\cos(\theta_k - \theta_l),\ U_6 = -\sum_{k=n+2}^{n+m+1} V_{kk}^2 B_{kk}$$

$$U_7 = -\sum_{k=n+2}^{n+m+1}\int_{V_k^0}^{V_k}\frac{Q_k^d(V_k)}{V_k}\,dV_k,\ U_8 = -\frac{1}{2}\sum_{i=1}^{n+1} B_{ii} + \frac{x_{di}' - x_{qi}}{2x_{qi}x_{di}'}\left[\cos[2(\delta_i - \theta_i)] - 1\right]V_i^2$$

$$U_9 = -\sum_{i=1}^{n+1}\frac{1}{x_{qi}'}E_{qi}'V_i\cos(\delta_i - \theta_i),\ U_{10} = \frac{1}{2}\sum_{i=1}^{n+1} E_{qi}'^2\frac{x_{di}}{x_{di}'(x_{di} - x_{di}')}$$

$$U_{11} = \sum_{i=1}^{n}\frac{1}{2x_{qi}'}\left(E_{di}' + 2E_{di}'V_i\sin(\delta_i - \theta_i)\right),\ U_{12} = -\sum_{i=1}^{n}\frac{V_i^2}{2x_{qi}'},\ U_{13} = \sum_{i=1}^{n}\frac{E_{di}'^2}{2(x_{qi} - x_{qi}')}$$

$$U_{14} = -\sum_{k=n+2}^{n+m+1} V_k I_{Lk}\sin(\theta_i - \phi_i).$$

We next express the overall dynamic equations in terms of partial derivatives of $U(\delta, \theta, V, E_q', E_d', I_L)$. This expression can facilitate the construction of energy functions:

$$
\begin{aligned}
0 &= -\frac{\partial U}{\partial \phi_i}\left(\delta, \theta, V, E_q', E_d', I_L\right) \\
0 &= -V_i \frac{\partial U}{\partial V_i}\left(\delta, \theta, V, E_q', E_d', I_L\right) \\
0 &= -\frac{\partial U}{\partial \phi_k}\left(\delta, \theta, V, E_q', E_d', I_L\right) \\
0 &= -V_i \frac{\partial U}{\partial V_k}\left(\delta, \theta, V, E_q', E_d', I_L\right) \\
\dot{\delta}_i &= \omega_i \\
M_i \dot{\omega}_i &= -D_i \omega_i - \frac{\partial U}{\partial \delta_i}\left(\delta, \theta, V, E_q', E_d', I_L\right) \\
T_{doi} \dot{E}_{qi}' &= E_{fdi} - \left(x_{di} - x_{di}'\right) \frac{\partial U}{\partial E_{qi}'}\left(\delta, \theta, V, E_q', E_d', I_L\right) \\
T_{qoi} \dot{E}_{di} &= -\left(x_{qi} - x_{qi}'\right) \frac{\partial U}{\partial E_{di}'}\left(\delta, \theta, V, E_q', E_d', I_L\right) \\
T_{vi} \dot{E}_{fdi} &= -E_{fdi} - \mu_i k_i V_i \cos\left(\theta_i - \phi_i\right) + l_i.
\end{aligned}
$$

In the preceding set of equations, only the last equation does not contain the analytical expression of the partial derivative of $U\left(\delta, \theta, V, E_q', E_d', I_L\right)$. We next add terms into $U\left(\delta, \theta, V, E_q', E_d', I_L\right)$ in order to make every equation contain the partial derivative of U. Note that

$$
\begin{aligned}
\frac{\partial U}{\partial E_{qi}'}\left(\delta, \theta, V, E_q', E_d', I_L\right) &= \frac{x_{di} E_{qi}'}{x_{di}'\left(x_{di} - x_{di}'\right)} - \frac{V_i \cos\left(\theta_i - \phi_i\right)}{x_{di}'} \\
&= \frac{-1}{x_{di} - x_{di}'}\left(T_{doi} \dot{E}_{qi}' - E_{fdi}\right).
\end{aligned} \tag{6.12}
$$

We next define two extra terms:

$$
\begin{aligned}
U_{15} &= -\frac{1}{2} E^T B^{-1} A E \\
U_{16} &= -C^T E,
\end{aligned} \tag{6.13}
$$

where

$$
\begin{aligned}
E &= [E_1, \ldots, E_n], \\
T &= \text{blockdiag}[T_1, \ldots, T_n], \\
A &= \text{blockdiag}[A_1, \ldots, A_n], \\
B &= \text{blockdiag}[B_1, \ldots, B_n], \\
C &= [C_1, \ldots, C_n]^T,
\end{aligned}
$$

$$E_i = [E_{qi}, E_{fdi}]^T,\ T_i = \begin{bmatrix} T'_{doi} & 0 \\ 0 & T'_{vi} \end{bmatrix},\ A_i = \begin{bmatrix} 0 & 1 \\ -\dfrac{\mu_i k_i x'_{di}}{x_{di} - x'_{di}} & -1 \end{bmatrix},\ B_i = \begin{bmatrix} \dfrac{x_{di} - x'_{di}}{x'_{di}} & \alpha_i \\ -\mu_i k_i & \beta_i \end{bmatrix},$$

and

$$C_i = [0, l_i]^T.$$

It follows that

$$\frac{\partial U}{\partial E} = \frac{\partial U}{\partial E} - B^{-1}AE - B^{-1}C, \tag{6.14}$$

and the voltage dynamics E can be written as the following equation:

$$T\dot{E} = -B\frac{\partial U}{\partial E}.$$

We next add these two terms into Equation 6.11:

$$U\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right) = U\left(\delta, \theta, V, E'_q, E'_d, I_L\right) + U_{15} + U_{16} \tag{6.15}$$

The overall system can thus be rewritten as the following compact set:

$$\begin{aligned}
0 &= -\frac{\partial U}{\partial \phi_i}\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right) \\
0 &= -\frac{\partial U}{\partial \phi_k}\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right) \\
0 &= -V_i\frac{\partial U}{\partial V_i}\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right) \\
0 &= -V_k\frac{\partial U}{\partial V_k}\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right) \\
\dot{\delta}_i &= \omega_i \\
M_i\dot{\omega}_i &= -D_i\omega_i - \frac{\partial U}{\partial \delta_i}\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right) \\
T'_{qoi}\dot{E}'_{di} &= -\left(x_{qi} - x'_{qi}\right)\frac{\partial U}{\partial E'_{di}}\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right) \\
T\dot{E} &= -B\frac{\partial U}{\partial E}\left(\delta, \theta, V, E'_q, E'_d, E_{fd}, I_L\right).
\end{aligned} \tag{6.16}$$

Clearly, the new model can be written as the compact form of Equation 6.3, where

$$u = [\theta, \phi]^T,\ w = \log V,\ x = \left[E'_d, E'_q, E_{fd}\right],\ y = \delta,\ z = \omega, \text{ and}$$

$$T = \text{blockdiag}\left[I_n, \frac{T'_{q0}}{x_q - x'_q}, B^{-1}T\right].$$

We next summarize a key result for constructing the above network-preserving transient stability model.

Theorem 6.2: Existence of the Analytical Energy Function (Chu and Chiang, 2005)

Consider the new class of network-preserving power system models described in Equations 6.8–6.12. Define

$$\begin{aligned} W\left(\omega, \delta, \theta, V, E_q', E_d', I_L, E_{fd}\right) &= k(\omega) + U\left(\delta, \theta, V, E_q', E_d', I_L, E_{fd}\right) \\ &= \frac{1}{2}\omega^T M\omega + \sum_{i=1}^{16} U_i, \end{aligned} \tag{6.17}$$

where U_i are defined in Equations 6.11 and 6.13, respectively. Then, $W\left(\omega, \delta, \theta, V, E_q', E_d', I_L, E_{fd}\right)$ is an (analytical) energy function for the new class of network-preserving power system models.

A physical interpretation of each term of the energy function (Eq. 6.17) along with a comparison with other energy functions proposed for different transient stability models, such as the models proposed in BH (Bergen and Hill, 1981), the NM model (Narasimhamurthi and Musavi, 1984), and the TAV model (Tsolas et al., 1985), are summarized in Table 6.1.

6.4 NONEXISTENCE OF ENERGY FUNCTIONS FOR LOSSY MODELS

In this section, the nonexistence of a general energy function for power systems with losses will be shown. While this may be a disappointing message, it is also shown that, under certain conditions, there is an energy function (defined in a compact set of the state space) that can be used in direct methods for the stability analysis of power systems with a certain degree of losses. One implication of these results is that any general procedure attempting to construct an energy function for a power system with losses must include a step that serves to check for the existence of an energy function.

Consider a lossy classical power system stability model. Let the loads be modeled as constant impedances. The dynamics of the ith generator can be represented by the following equations:

$$\begin{aligned} \dot{\delta}_{in} &= \omega_i - \omega_n \\ M_i\dot{\omega}_i &= P_i - D_i\omega_i - \sum_{j\neq i, j=1}^{n} E_iE_jB_{ij}\sin\left(\delta_i - \delta_j\right) - \sum_{j\neq i, j=1}^{n} E_iE_jG_{ij}\cos\left(\delta_i - \delta_j\right), \end{aligned} \tag{6.18}$$

where node n serves as the reference node. E_i is the constant voltage behind direct axis transient reactance and M_i is the generator's moment of inertia. The damping

Table 6.1 Various Network-Preserving Power System Models and Their Corresponding Potential Energy Functions

Potential energy	Formula	BH model	NM model	TAV model	CC model
Mechanical power injections	U_1	Y	Y	Y	Y
Transfer between generator terminal buses	U_2	Y	Y	Y	Y
Transfer between generator terminal buses and load buses	U_3	Y	Y	Y	Y
Active power demand at load buses	U_4	Y	Y	Y	Y
Transfer between load buses	U_5	Y	Y	Y	Y
Loss at each load bus	U_6		Y	Y	Y
Reactive demand at load buses	U_7		Y	Y	Y
Transfer between generator internal buses and terminal buses	U_8			Y	Y
Quadrature axis voltage variations at generator internal buses	U_9			Y	Y
Quadrature axis voltage variations at generator internal buses actions of exciters at generator internal buses	U_{10}			Y	Y
D-axis voltage variations	U_{11}				Y
Damper winding on the Q-axis	U_{12}				Y
Damper winding on the Q-axis	U_{13}			Y	Y
Constant current loads	U_{14}				Y
Interactions between Q-axis voltage variations and exciters	U_{15}				Y
Exciters at generator internal buses	U_{16}				Y

CC, Chu and Chiang.

constant D_i is assumed to be positive. The G_{ij} term represents the transfer conductance of the i–j element in the reduced admittance matrix of the system (Eq. 6.18). $P_i = P_{mi} - E_i^2 G_{ii}$, where P_{mi} is the mechanic power. Furthermore, when uniform mechanical damping is assumed, that is, $D_i/M_i = C$, $i = 1, 2, \ldots, n$, the dynamics of the ith generator can be represented by the following equations:

$$\begin{aligned}\dot{\delta}_{in} &= \omega_{in}, i = 1, 2, \ldots, n-1 \\ \dot{\omega}_{in} &= (P_{mi} - P_{ei})M_i^{-1} + (P_{mn} - P_{en})M_n^{-1} - C\omega_{in} \\ &= -C\omega_{in} + f_{in}(\delta).\end{aligned} \tag{6.19}$$

We first consider the following system, which describes one machine connected to an infinite bus through a lossy transmission line:

$$\begin{aligned}\dot{\delta} &= \omega \\ M\dot{\omega} &= P - D\omega - EB\sin(\delta) - EG\cos(\delta),\end{aligned} \tag{6.20}$$

where δ is the phase angle of the machine with respect to the infinite bus. Consider the following function for (Eq. 6.20):

$$V(\delta, \omega) = \frac{1}{2}M\omega^2 - P\delta - EB\cos(\delta) + EG\sin(\delta). \tag{6.21}$$

The derivative of Equation 6.21 along the trajectory of the system (Eq. 6.20) is given by

$$\dot{V}(\delta, \omega) = -D\omega^2 \le 0.$$

Hence, the function (Eq. 6.21) is a general energy function for the one-machine infinite bus system (Eq. 6.20).

The extension of the energy function (Eq. 6.21) to multimachine power systems represents a great challenge. In their pioneer paper (Abiad and Nagappan, 1966), El-Abiad and Nagappan proposed the following function for the multimachine power systems with losses:

$$\begin{aligned} V_1(\delta, \omega) = &\frac{1}{2}\sum_{i=1}^{n} M_i\omega_i^2 - \sum_{i=1}^{n} P_i(\delta_i - \delta_j^s) \\ &- \sum_{i=1}^{n}\sum_{j=i+1}^{n+1} E_iE_jB_{ij}\left[\cos(\delta_i - \delta_j) - \cos(\delta_i^s - \delta_j^s)\right] \\ &- \sum_{i=1}^{n}\sum_{j=i+1}^{n+1} E_iE_jG_{ij}\left[\sin(\delta_i - \delta_j) - \sin(\delta_i^s - \delta_j^s)\right]. \end{aligned} \tag{6.22}$$

Differentiating the function V_1 (δ, ω) along the trajectories of the system (Eq. 6.18) gives

$$\begin{aligned} \dot{V}_1(\delta, \omega) &= \frac{\partial V}{\partial \delta}\dot{\delta} + \frac{\partial V}{\partial \omega}\dot{\omega} \\ &= -\sum_{i=1}^{n} D_i\omega_i^2 - 2\sum_{i=1}^{n}\sum_{j=i+1}^{n+1} E_iE_jG_{ij}\omega_i\cos(\delta_i - \delta_j). \end{aligned} \tag{6.23}$$

This function V_1 (δ, ω) is therefore not an energy function for the system (Eq. 6.18). Note that $\dot{V}_1$ will be negatively definite for large values of ω_i.

Many attempts have since been made to derive general energy functions for the system (Eq. 6.18) but without success. Another approach pursued by Uemura et al. (1972), Athay et al. (1979), Kakimoto et al. (1978), and Guindi and Mansour (1982) is based on numerical approximations accommodating the transfer conductance effects. It should be stressed that these approximations lead to functions that do not possess the properties of analytical energy functions.

At present, there is no general energy function for multimachine lossy stability models. The following key result has been shown in (Chiang, 1989).

Theorem 6.3: Nonexistence of an Energy Function

There is no general energy function for the power system (Eq. 6.18) with losses, where $n \ge 2$ is the number of generators.

Remarks:

1. This main result shows that there is no general analytical energy function for multimachine power systems with losses (Eq. 6.18). The usefulness of existing local, analytical energy functions in direct methods is not clear. It should be noted that in order to be useful, these local energy functions must be at least well-defined in the stability region of the postfault system.
2. For the simple system of one-machine infinite bus through a lossy transmission line, it can be shown (by direct analysis of the associated differential equation) that there is no limit cycle in the state space. The existence of a global analytical energy function (e.g., Eq. 6.21) for these one-machine infinite bus systems also confirms this point.
3. One key implication is that any general procedure attempting to construct an analytical energy function must include a step serving to check for the existence of an analytical energy function. This step essentially plays the same role that the Lyapunov equation plays in determining the stability of an equilibrium point.

6.5 EXISTENCE OF LOCAL ENERGY FUNCTIONS

We will show the existence of an energy function defined on any compact set of the state space of the system (Eq. 6.19) with small losses. It is well known that, under the assumption that the transfer conductances of the system (Eq. 6.19) are zero, there exists an energy function for this system. Indeed, with $G_{ij} = 0$, we define a function, $V(\delta, \omega)$, as follows:

$$\begin{aligned} V(\delta, \omega) = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \Bigg\{ & \frac{1}{2} M_i M_j \omega_{ij}^2 - (P_i M_j - P_j M_i)(\delta_{ij} - \delta_{ij}^s) \\ & + \left(\sum_{i}^{n} M_i \right) E_i E_j B_{ij} (\cos\delta_{ij} - \cos\delta_{ij}^s) \Bigg\} \\ = {} & V_k(\omega) + V_p(\delta). \end{aligned} \tag{6.24}$$

At this point, some concepts from the mathematical dynamical system theory are recalled. Let $\Phi^f(.)$ (resp. $\Phi^g(.)$) be the flow of a vector field f (resp. g) and $\Omega\ (f)$ (resp. $\Omega\ (g)$) be its nonwandering set. We say f is Ω-stable if, for every small C^r-close ($r > 0$) vector field of f (i.e., every vector field close to f in the C^r topology), say, g, there exists a homeomorphism h: $\Omega\ (f) \to \Omega\ (g)$, which is orbit preserving (Nitecki and Shub, 1975; Pugh and Shub, 1970). The next analytical result shows that the system (Eq. 6.19) with transfer conductance $G = 0$ is Ω-stable.

Theorem 6.4: Ω-Stable

If the system (Eq. 6.19) with $G = 0$ satisfies the assumption of hyperbolic equilibrium points, then this system is Ω-stable.

Theorem 6.4 implies that, under the assumption of hyperbolicity for the power system (Eq. 6.19) without losses, there exists a positive number, α, such that if the transfer conductance of the system (Eq. 6.19) satisfies $|G| < \alpha$, then the nonwandering set of the system (Eq. 6.19) with a certain transfer conductance also consists entirely of equilibrium points. This property, in combination with an analytical result regarding the fine filtration (Nitecki and Shub, 1975), leads to the following result: for any compact set S of the state space of the system (Eq. 6.19), there exists a positive number, α, such that if the transfer conductance of the system (Eq. 6.19) satisfies $|G| < \alpha$, then there exists an energy function defined on this compact set S.

The above analytical results confirm the current practice of using numerical energy functions in the area of direct methods. The issues, however, are the reliability and the performance of these numerical energy functions in stability analysis. Further investigation is needed, although these numerical energy functions seem to have performed very well in direct methods.

6.6 CONCLUDING REMARKS

The existence of analytical energy functions for lossless power system transient stability models, including the network-reduction and network-preserving models, has been shown. A methodology for deriving the analytical energy functions has been presented. Physical explanations of each term in the derived analytical energy functions have been provided.

The nonexistence of general analytical energy functions for lossy power system transient stability models has been shown. One key implication is that any general procedure attempting to construct an energy function for a lossy power system transient stability model must include a step that checks for the existence of an energy function. This step essentially plays the same role as the Lyapunov equation in determining the stability of an equilibrium point.

The existence of a local analytical energy function defined on any compact set of the state space of a power system transient stability model with small losses has been shown. This analytical result is local, and it is not clear whether a local analytical energy function is useful or not. The analytical results derived in this chapter confirm the current practice of using numerical energy functions in the area of direct methods. The topic of how to construct numerical energy functions will be discussed in the next chapter.